

Tripoli Level 3 Certification Report

SA Great

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1 Project Overview

The purpose of this document is to provide the Tripoli Technical Advisory Panel (TAP) members with the required information pertaining to the design, manufacturing, and preparation of Tripoli Rocketry Association (TRA) Member 32226 Level 3 Certification rocket, hereafter referred to as SA Great.

SA Great is being prepared and constructed by Lachlan Miegel for the purpose of achieving a TRA Level 3 HPR Certification. This rocket will be entirely designed, manufactured, and prepared independently using minimal commercial off-the-shelf (COTS) components where possible.

1.1 Proposed Rocket

The proposed rocket has been initially developed using OpenRocket (OR), used for conceptualisation and simulations, with OnShape being used for development for the physical model.

SA Great is designed to have a continuous 114mm (4.5") ID airframe, from COTS carbon tubes, standing at 2550 mm tall with a 98 mm motor mount and four-fin design. The proposed rocket is intended to reach 25 000ft at Mach 2.0, implementing a dual separation dual deploy recovery mechanism. The airframe, bar body tubes, is intended to be scratch built, with COTS avionics systems being used as the primary and secondary flight computers, with COTS GPS.

1.2 Proposed Motor

The motor proposed is an AeroTech N3300R, using the RMS-98/15360 casing borrowed from ARES Rocketry. This motor will propel the rocket to Mach 2, with the airframe undergoing an estimated 28 G-forces during motor burn. The motor specifications is shown below in Table 1.

Table 1: AeroTech N3300R Motor Specifications

Descriptor	Specification
Total Impulse	13 410 N s
Burn Time	4.1 s
Peak Thrust	4 226 N
Propellant Weight	7 338 g
Total Weight	12 054 g

2 Design & Manufacturing Details

2.1 Airframe & Simulation

Pictured in Figure 1 is the OpenRocket of SA Great, showing all key features of the airframe including approximate CG and CP markers. Figure 2 depicts the simulation of the rocket completed using OpenRocket.

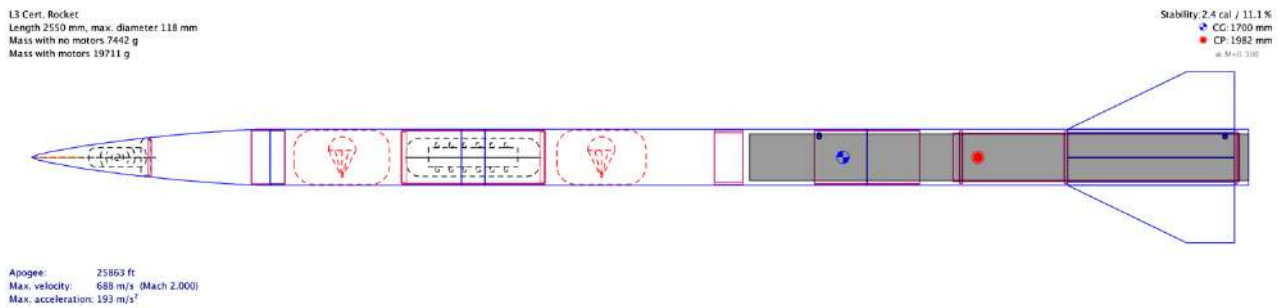


Figure 1: OpenRocket simulation of SA Great showing the overall rocket configuration

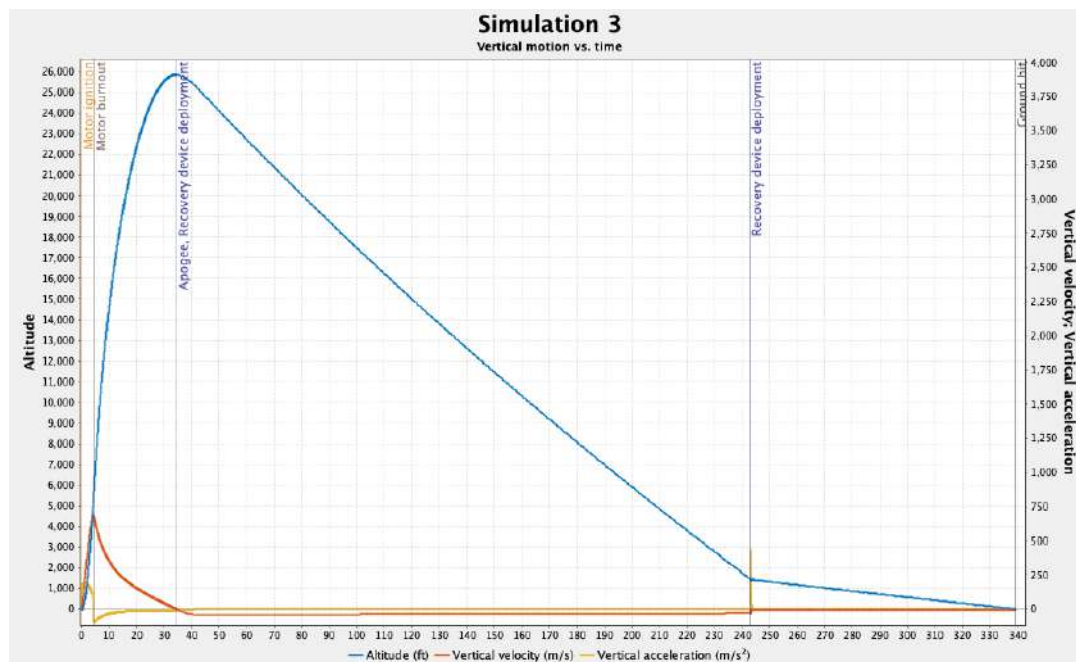


Figure 2: OpenRocket analysis output for SA Great

2.1.1 Booster Tube

The airframe is of composite construction, featuring roll-wrapped carbon tubes, carbon plate fins and centering rings, carbon motor tube, fibreglass coupler and nosecone, and 6061-T6 aluminium bulkheads.

The booster tube will be comprised of two 800mm long segments of tube that will be bonded together. Purchasing a body tube at the desired 1600mm long segment would require high shipping costs, so two 800mm long segments is more cost effective. The fins will be waterjet cut from 5mm thick carbon plate, then chamfered into a double-diamond profile. The fins will be bonded onto the motor tube, made in-house, using Epikote MGS BPR 135 bonding paste, using an MDF jig for alignment. Waterjet cut centering rings will be made from the same 5mm carbon plate and bonded at equidistant spacing on the motor tube using the same paste.

The body tube will have fin slots cut into the tube and then the fin-can slid and bonded to the body tube with large structural epoxy fillets of approximately 10mm radius, using the same Epikote epoxy. Finally, a tip-to-tip over the fins will complete the booster tube, adding approximately 0.8mm of thickness at the fin top chords, and 1.2-1.6mm of thickness at the root chord. Fin flutter calculations

demonstrated the flutter velocity to be about Mach 2.6.

Pictured below in Figure 3 is how the fins will appear after the structural fillets are applied. This picture is indicative only of the filleting style and epoxy used.



Figure 3: Indicative fin geometry and structural fillet design

The booster tube will also house the motor retainer, using a 3/8" threaded hole to fasten the top enclosure, with the motor retainer being radially attached. This part will be machined out of 6061-T6 aluminium and also include a mounting point for a U-bolt. Thrust will go through the airframe via the aft enclosure interfacing with the motor tube, rather than through the motor retainer.

Pictured below in Figure 4 is the aft section of the airframe, whilst the motor is not pictured, the thrust will be transferred through the airframe via the beige ring through the lip on the motor casing. The beige ring represents the aft centering ring made from carbon fibre, which is bonded to the motor tube, trailing fin edge, and booster body tube using Epikote. This will deliver a maximum force of approximately 4300N. The motor is then held into the airframe by a retainer that uses the 3/8" threaded hole at the top enclosure.

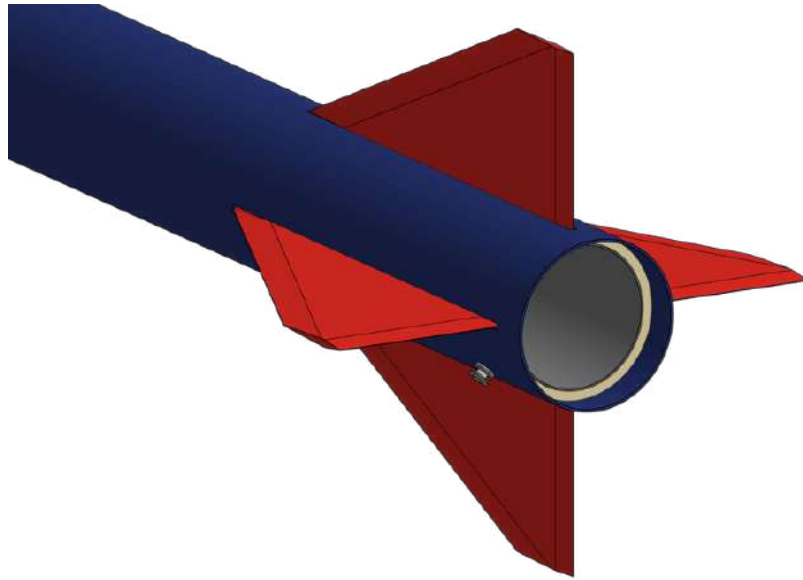


Figure 4: Aft section of the airframe showing the motor retainer and thrust transfer interface

Figure 5 below is the motor retainer with the body tube made transparent for visualisation. 12 M5 countersunk screws will secure the motor retainer into the body tube, preventing the motor from sliding free from the airframe after motor burnout. The retainer will need to support the mass of the motor whilst on the pad, and with the 12 bolt design, will be able to hold approximately 80kg, using the known shear strength of the bolt and estimates on what the composite tubes can support.



Figure 5: CAD model of the motor retainer assembly

2.1.2 Nosecone & Extension Tube

The nosecone construction will use a method that involves a singular female mould that is exactly one half of a nosecone. This mould will be 3D printed and then prepared to create one half of the nosecone shell. Two halves of the shell will be produced and then bonded internally with fibreglass strips. The fibreglass construction is chosen over carbon fibre due to the GPS being located in the

nosecone, allowing for RF transparency. The nose tip will be turned out of brass, acting as ballast, and providing an aerodynamic point to withstand the intense speeds.

The nosecone will house a bulkhead for the GPS to sit, and an attachment point for a shock cord, in a similar way to previous ARES rockets. This bulkhead will have a ring that epoxies into the nosecone shell, with the bulkhead fixturing the GPS and U-bolt. The epoxy used to bond the ring to the inside of the nosecone is Epikote MGS-BPR 135 bonding paste, with a shear strength of $20\,000\,000\text{ N m}^{-2}$ on a 1 mm^2 sample, as such should withstand the shock load from a parachute release. To further hold the GPS bulkhead in place, the nosetip will sit on the shoulder of the nosecone shell and tension the bulkhead via an M6 threaded rod, adding a secondary fixturing method of the parachute attachment point. This threaded rod will screw into the nosetip and bolt onto the bulkhead, using Loctite.

The nosecone will have a short fibreglass coupler that has an aluminium ring that allows the nosecone to be affixed to the extension tube using radial bolts, with the separating point located at the avionics bay coupler. The overall expected mass of the nosecone and internal components is approximately 900 grams. This sub-assembly is seen below in Figure 6.

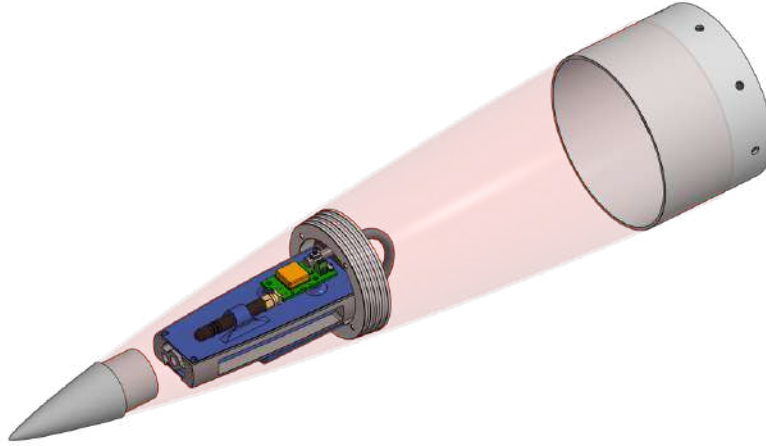


Figure 6: Nosecone assembly including internal GPS bulkhead and attachment features

2.1.3 Coupler & Switchband

The coupler will be a 300mm long fibreglass construction, manufactured by laying fibreglass internally to a prepared body tube for a precision fit. Two bulkheads will enclose the avionics bay with the fibreglass tube protecting the avionics, machined out of 6061-T6 aluminium. The avionics bay will be 3D printed out of ABS-GF, a proprietary ABS blend that has incorporated glass fibres for added structural and thermal properties. The avionics bulkheads will have a mounting space for a U-bolt and the threaded rods securing the two bulkheads and providing tension.

The avionics bay is secured to the bulkheads with three threaded rods that run through the sled and bolt to either bulkhead. There is a large central M8 threaded rod that will resist the tension placed by the two parachute fixtures, and two M4 threaded rods that resist the sled from spinning between the bulkheads and providing auxiliary tension strength. This design ensures the avionics bay cannot

spin between the bulkheads and has ample strength to resist the shock force during separation and recovery events.

The avionics sleds (light blue) are secured to the chassis using M3 screws into heat-insert nuts placed into the chassis (dark grey), with Loctite to ensure they do not come loose during flight. Each component is then secured to the sleds via more M3 screws into heat-insert nuts placed into the sleds, allowing for a modular design that is secure.

The switchband will be cut from a section of body tube, 50mm in length and epoxied centrally to the coupler. The coupler is designed to be of sliding nature and let the booster and extension tubes slide freely without incurring any bending moment. The coupler also contains a number of holes (3x 5mm and 1x 30mm) for switch functionality and GoPro vision, also acting as pressure equaliser holes such that the BR's take accurate measurements.

Figure 7 below is the coupler and avionics bay, with the two flight computers (righthand side), 9V battery mounts (lefthand side), and switch mount (dark blue) all visible.

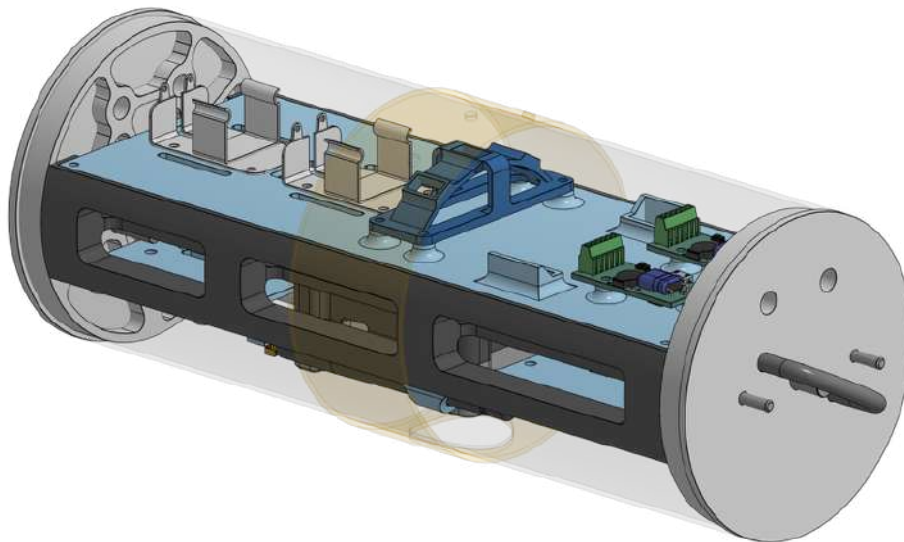


Figure 7: Avionics bay and coupler assembly

Figure 8 below is a closeup of the bulkhead, the three threaded rods are seen on the end of the bulkhead, with the two vacant holes above there for wire passthrough for ejection charges. Nylock nuts with a washer and Loctite shall be used to secure all the threaded rods.

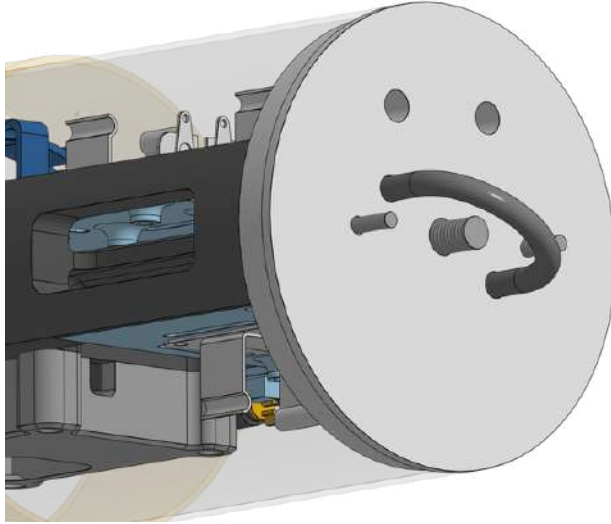


Figure 8: Avionics bulkhead showing threaded rod and wiring pass-through features

2.2 Avionics

The avionics system will feature two entirely independent flight computer systems, a Blue Raven and an RRC3+. These will control the main and apogee separation events, and their respective redundant separation events. The GPS will use a Featherweight system, housed in the nosecone for RF transparency. The FC's will use 9v batteries and slide switches for arming, accessed through the switch band. The GPS will use a Li-Po cell, armed via a magnetic switch in the nosecone. The pyrotechnics will be pre-assembled and attached to the avionics bulkhead connectors using Wago snap connectors on launch-day, then armed via the switches accessed through the switchband. The schematic is seen in Figure 9

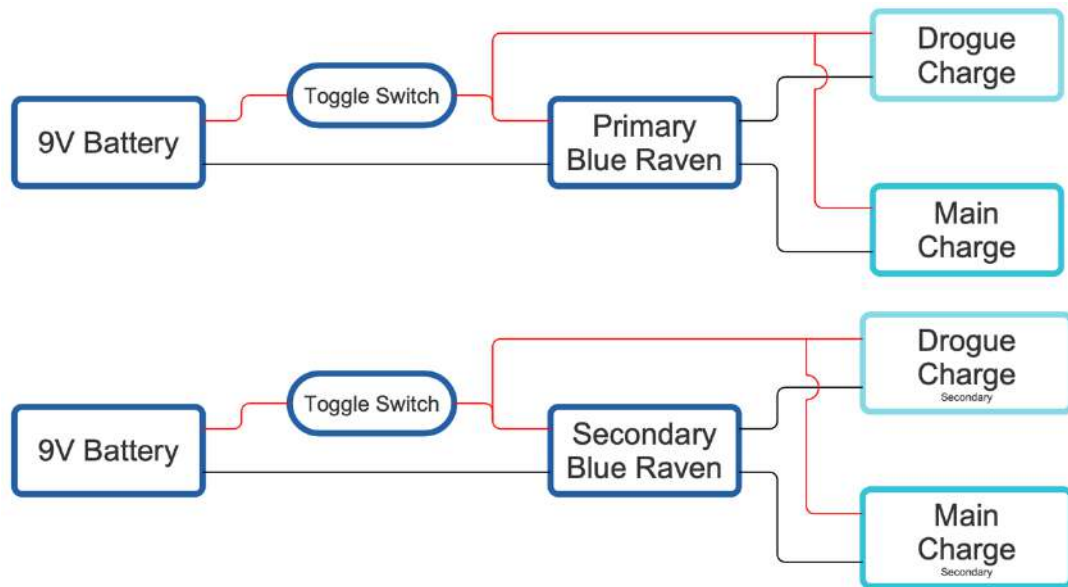


Figure 9: Flight avionics schematic showing redundant flight computer systems

Each FC will use the Main and Apogee output, with the secondary RRC3+ programmed to trigger

0.5-1 second after the primary BR signals ejection. Each end of the avionics bulkhead contains 3 snap connectors that are epoxied on using Epikote MGS BPR 135, the negative (black) lead from each BR will go into its own connector, and the positive (red) lead will go to the third connector. Each charge has two wires, one wire will go into a connector with its required negative lead, and the other wire will share the common positive connector. This will be mirrored on both avionics bulkheads.

A separate system of ARES Rocketry avionics will also fly, housed on a separate side of the avionics sled, it will operate as a purely data-gathering system and have no effect on the flight-critical avionics, and hence is not depicted in the schematic.

Finally, a GoPro will also fly within the avionics bay, armed via turning the device on at the pad, it will record the duration of the flight and is not controlled by any additional hardware.

2.3 Recovery

The recovery employs a dual separation dual deploy mechanism, with the first event at apogee deploying a 24' pilot chute, and second event occurring at approximately 1500 ft with a 72" Fruity Chutes Compact Iris. The separation will occur using 4F black powder (BP) packed into nylon burst tube, a proven method thanks to ARES that requires a few grams of BP to induce separation. A calculated 1.2 grams of BP is required as described in Appendix A, although this will be increased by 0.5 grams and validated through ground-testing. The calculations for shear pins are seen in Appendix B, requiring three shear pins per bay, although this number will be validated through ground separation testing. More pins may be required to secure the main parachute bay to ensure premature separation does not occur prior to main deployment during decent. There will be one vent hole in each parachute bay, 3mm in size and placed such that the recovery system will not obstruct air bleed to ensure pressure equalisation.

Below in Figure 10 an example of the nylon burst tube containing BP, where these sealed vials contain an e-match and have their own atmosphere allowing for easier ignition at altitude. These tubes will be wrapped in a Nomex blanket and secured to the internal wall of the rocket using tape to reduce the risk of movement during flight. The e-match will be secured into the connectors as described above.



Figure 10: Example nylon burst tube assembly used for separation charges

All lines are 4mm braided SK-75 Dyneema, an aramid fibre with approximately 2700kg of force before yielding. The lines shall be finger trapped and attached to each M6 316 stainless steel U-bolts using soft shackles at the nosecone, integrated avionics coupler, and motor retainer, with the line diagram pictured in Figure 11.

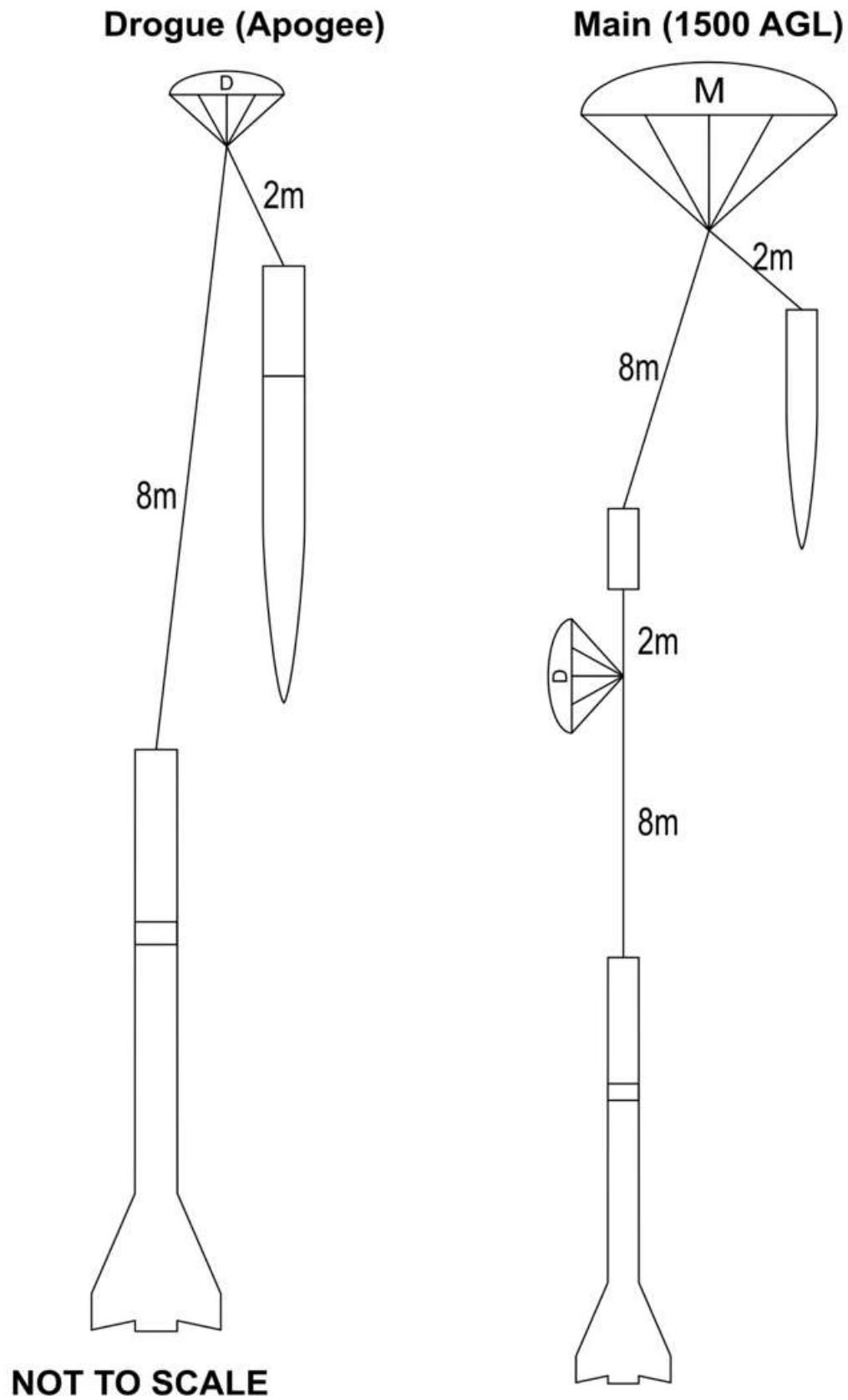


Figure 11: Recovery system line attachment configuration

3 Manufacturing

3.1 Airframe Component

To begin the manufacturing process, the nosecone was the first component to be produced. The first step was manufacturing the single split mould, done via 3D printing in two halves on a BambuLab P1S, glued together with two-part 5 minute epoxy and locating dowels. The moulds were sprayed with Dulux Autoshield Acrylic Filler Primer and sanded up to 600 grit to smooth out the printer layer lines. After the mould was prepared, wax and PVA were applied in several coats to ensure the composite part releases after curing. The mould is seen below in Figure 12.



Figure 12: Nosecone mould after priming

The next step for the nosecone was layup, involving wet laying fibreglass into the nosecone mould, then pulling a vacuum to create each half. Figure 15 below shows one half of the nosecone after demoulding, with the mould ready for the next layup. This process was repeated to form the two halves, which were then placed back into the mould and the dog ear tabs cut off.



Figure 13: Nosecone half after demoulding

After the two halves were demoulded and trimmed, the next step was bonding the two halves together, using 3D printed alignment rings, and strips of fibreglass wet laid on the internal surface. The resulting nosecone is of sturdy construction, requiring minimal postprocessing to prepare for sanding. The nosecone with the jigs attached is seen in Figure ??, with the final shell seen in Figure ??.



Figure 14: Nosecone with 3D printed jigs attached for final glue-up



Figure 15: Final nosecone shell

After the nosecone was completed, the two couplers were created, using a method of wrapping one of the body tubes in Mylar, then wet laying composite on the internal surface to create a snug-fitting coupler. The Mylar is 125 microns thick, making the couplers slide with minimal friction. The final component to be created was the motor tube, using a 3D printed mandrel and Mylars as the release film, carbon fibre was wet laid and left to air cure, then trimmed to length. These processes completed the airframe tube elements, as shown in Figure 16.



Figure 16: Tube components of the airframe, switchband seen in foreground

After all the airframe tube components were completed, attention was turned to the fin can / booster tube. The process started by cutting out the fins from 5 mm carbon fibre plate, and hand chamfered using a belt sander to achieve a double diamond profile. From here, a jig was lasercut from 6 mm MDF that could hold all four fins to the motor tube for the first glue up, as shown in Figure 17.



Figure 17: Fin can glueup

After the fins were tacked onto the motor tube with Epikote MGS BPR 135 bonding paste, centering rings were bonded on to the motor tube using the same paste. From here, the fin can was slid into the booster tube with slots cut in the aft end, and the fins were tacked into place with the same bonding paste, as shown in Figure 18.



Figure 18: Fin can glueup

Before tip to tip, the final structural fillets were completed, using the same bonding paste, at a 25 mm radius. These fillets are seen below in Figure 19.



Figure 19: Fin can glueup

The tip to tip used four staggered layers prioritising fin root chord strength to reduce help reduce flutter. The carbon fibre was 200gsm twill, layered quasi-isotropically for maximum stiffness, wet laid and vacuum bagged using Westlake L285 laminating resin. The tip to tip fresh out the bag is seen in Figure 20.



Figure 20: Tip to tip fresh out of the bag

Before painting, the fin can required a large amount of preparation, using body filler to smooth out any pinholes and imperfections for a smooth finish. Additionally, the fin fillet leading and trailing edges were built up using filler to create a seamless fin can. The body filled booster tube is seen in Figure 21.



Figure 21: Bogged up fin can

Completing the fin can closed out the manufacturing of the airframe, with the final airframe before painting seen in Figure 22.

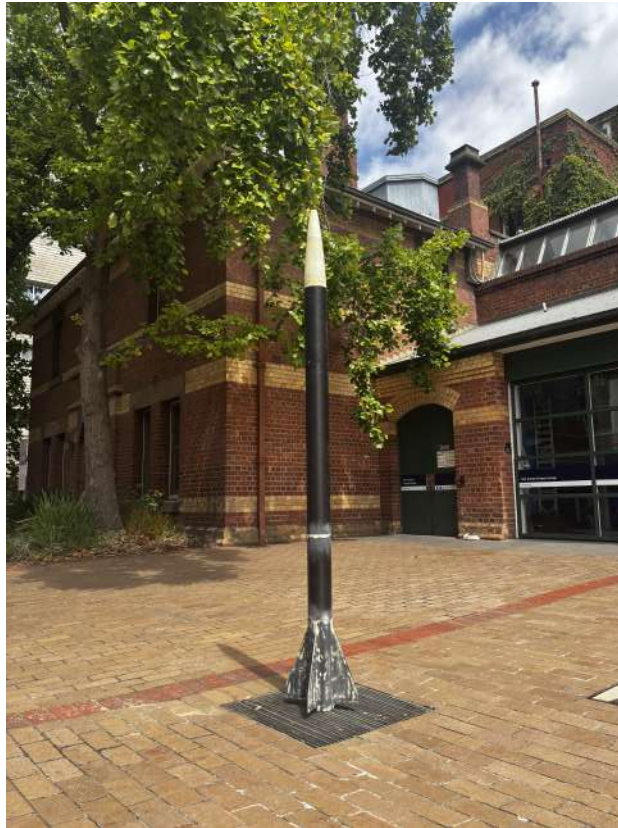


Figure 22: Completed airframe before finishing

3.2 Machining

There are a number of components within the airframe that are machined, primarily out of 6061-T6 aluminium. The first component worked on was the epoxy ring and GPS bulkhead located in the nosecone, and bonded into the fibreglass shell using the Epikote bonding paste. These were turned on a lathe, with the holes drilled and tapped afterwards. The final bulkhead is seen in Figure 23.



Figure 23: Nosecone epoxy ring and bulkhead

The next part to be machined was the two avionics bulkheads, completed on a Tormach 1100MX, and enclose the primary avionics bay in the centre of the rocket. These bulkheads also feature to fixture points for the parachute attachments, connected via M8 and M4 threaded rods that run through the avionics bay. The machining step is seen in Figure 24, with the final bulkheads shown in Figure 25.



Figure 24: First operation of avionics bulkhead on the Tormach



Figure 25: Completed avionics bulkhead

The two final parts were the motor retainer [not pictured], and the nosetip, which was turned out of brass by an external manufacturer. Aside from rail buttons, these comprise the final metallic sections of the rocket, enabling final assembly to progress.

3.3 Avionics Sub Assembly

After the bulkheads were completed, the avionics sleds and chassis were printed out of ABS-GF, and assembled using heat set nuts to mount all the required components. The GPS bay was the first one to be completed, as shown in Figure 26.



Figure 26: Featherweight GPS mounted onto nosecone bulkhead

The final Avionics Bay with the RRC3+ and Blue Raven is seen in Figure 27.

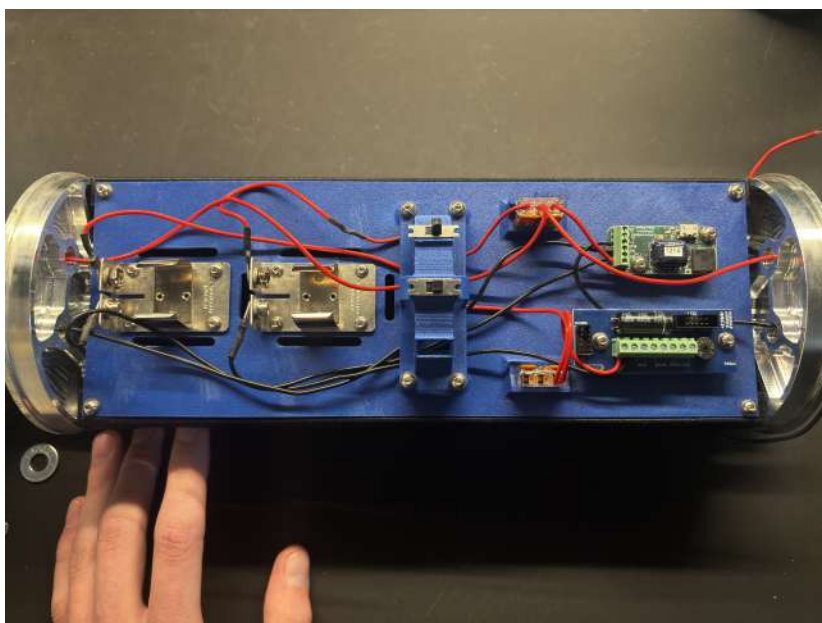


Figure 27: Avionics bay with components mounted, no batteries

3.4 Painting

The painting process took several days, comprising priming, base coat, colour coat, and final clear coat, involving several rounds of painting, wet sanding, and curing. The two paint colours each

required their own base coat, with the booster primed and tubes base coat applied in Figure 28.



Figure 28: SA Great in the paint booth, booster primed and base coats down

After priming was completed, the colour was put down, a candy finish, as shown in Figure 29.

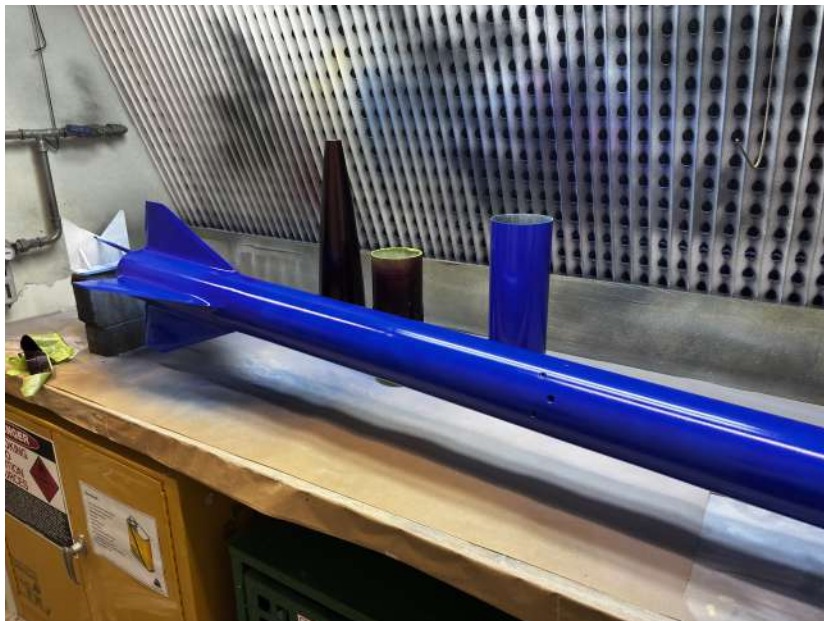


Figure 29: SA Great with colour applied

To finish off the painting process, the decals were applied, with 2-pack clear coat closing out the process, with some of these parts shown in Figure 30.



Figure 30: SA Great with decals applied to extension tube

This closed out the entire manufacturing process with the final rocket seen in Figure 31.



Figure 31: SA Great

4 Motor Amendment

As of July 2026, a new launch opportunity presented itself to attend BALLS 34 in Black Rock, Nevada. As such, motor availability and desire to have a successful flight has driven a change in motor selection to the **AeroTech M2050X-PS**, using ARES' 75/5120 Casing.

4.1 Updated Simulations

Under the new motor selection, the updated apogee is 11,800 ft, at Mach 1.2, with stability margin of 3.2. This stability will likely cause the rocket to weathercock, but has the added advantage of being stable on launch with the shorter 8' rails at BALLS 34.

The updated flight profile is seen in Figure 32.

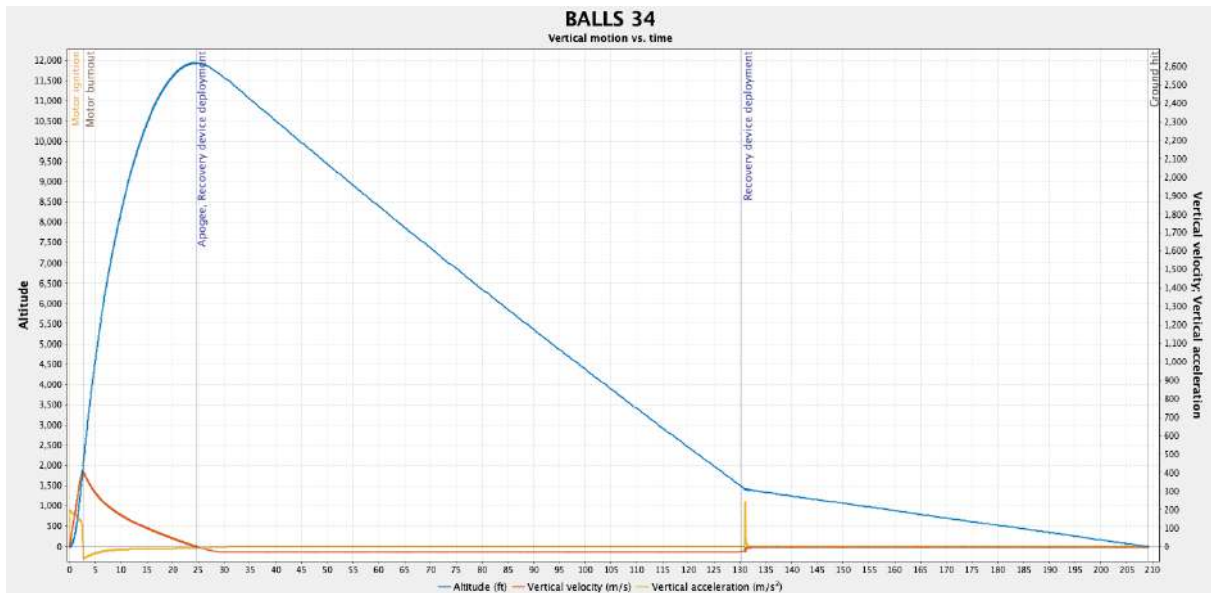


Figure 32: Flight simulation with M2050

4.2 Booster Tube Modification

In order to fit the new 75 mm motor, a custom mounting system has been devised in order to retain the motor and contain the thrust. There will be an extension bar going from the existing motor mount to the top of the 75/5120 casing, made out of 25 mm 6061-T6 Aluminium, and can withstand the buckling force that the M2050 will survive.

Another aspect of the retention method for the smaller motor are 3D printed centering rings that will clamp over the casing and keep the motor concentric within the airframe. The final aspect is a custom aeropack-style ring that will be machined from 6061-T6 and push the thrust from the lip of the motor into the existing aft centering ring, with a 3D printed version of this concept shown in Figure 33.



Figure 33: M2050 motor casing retained within airframe

A Black Powder Calculation

Calculated Form

Starting form. Basic calculated fields sample.

Body Tube Length (inches)

14.76

Body Tube Inner Diameter (Inches)

4.5

Pressure (8-15 psi)

10

Start low at 8. Ground test as you work your way up.

Below is the calculated BP charge. Remember to always ground test.

Force on Bulkhead (pounds)

158.9625

BP Charge (grams)

1.2

Figure 34: Black powder sizing calculation.

B Shear Pin Calculations

Pressure at apogee (25 000 ft) = 36 600 Pa

Shear pins – M3 Nylon, shear strength = 298.126 N

$$\Delta P = P_{\text{internal}} - P_{\text{external}} = 101.325 - 36.600 = 64.725 \text{ kPa}$$

Force due to pressure differential:

$$F = PA$$

$$F = 64725 \times 0.01021$$

$$F = 660.7 \text{ N}$$

Number of shear pins:

$$N = \frac{F}{F_{\text{pin}}}$$
$$N = \frac{660.7}{298.126} = 2.22 \approx 3 \text{ pins}$$

C Epoxies within Airframe

Epikote MGS BPR 135 Bonding Paste is used:

- Fin bonding onto motor tube
- Structural fin fillets
- Centering ring to motor tube and body tube
- GPS bulkhead onto nosecone internal
- Snap connector securing on avionics bulkheads

Westlake Systems L285 Laminating Resin is used:

- Motor tube manufacturing
- Switchband onto coupler
- Tip to Tip
- Nosecone